

Mayur Kishore Kumar

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SUMMARY

Machine Learning Engineer with experience building production ML and AI systems across semiconductor, healthcare, and financial domains. Experienced in predictive modeling, LLM applications, RAG, AI agents, and ML infrastructure, with hands-on experience across data engineering, model development, backend systems, and enterprise deployment.

SKILLS

Languages: Python, SQL, Java, C, C++

ML Infrastructure & Cloud: Linux, Docker, Kubernetes, MLflow, FastAPI, Flask, AWS, Azure, GCP, Git

Data Engineering & MLOps: Apache Spark, Airflow, PostgreSQL, TimescaleDB, ClickHouse

Generative AI & LLMs: RAG, AI Agents, MCP, LangChain, Embeddings, FAISS, ChromaDB, Ollama, Groq, n8n, LangGraph

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, XGBoost

Computer Vision: YOLO, OpenCV, Image Processing, PaddleOCR, EasyOCR, Tesseract

WORK EXPERIENCE

RAMBUS

Bangalore, India

Machine Learning Engineer Intern

Jul 2025 – Present

- Architected and deployed production ML and AI systems across semiconductor engineering workflows, spanning **predictive ML, Generative AI, RAG, agentic systems, MCPs, and AI developer tooling** in an **air-gapped RHEL/Linux environment**.
- Built an end-to-end **EDA resource prediction system** using IBM LSF and Spectre data, scaling the training dataset from **30K to 1.5M+ production jobs** across **4+ years**, evaluated multiple models and achieved **$R^2 \approx 0.87$** for 90th-percentile resource prediction using quantile regression.
- Designed and deployed an **enterprise RAG ecosystem** combining **BM25 keyword retrieval, vector search, reranking, and knowledge-graph retrieval**, with department-specific knowledge bases used across the organization and user feedback indicating **~40% average time savings** across AI-assisted workflows.
- Developed **~15 MCP integrations personally (~20 across the team)** for **Jira, Confluence, Jenkins, and ServiceNow** using Python/FastMCP, with the infrastructure **serving hundreds of users** and supporting **~30% of relevant ticket activity**.
- Engineered AI developer tooling and agentic workflows across **GitHub Copilot, Continue.dev, terminal-based agents, and OpenClaw**, including **~70 engineering skills and integrations**, deployed agent infrastructure supporting **~80–100 daily users** and **millions of tokens of usage per day**.
- Automated engineering workflows including **NetApp filesystem analytics, VS Code extension provisioning, and SFTP-based semiconductor data pipelines**, while developing **GenAI log-analysis and document-intelligence workflows** using LLMs, OCR, and enterprise APIs used by **20–30 finance users**.
- Supported enterprise AI adoption through **Level 1 and Level 2 training**, helping engineers integrate AI-assisted development and agentic workflows into day-to-day engineering tasks.

HUMANITARIANS AI

Boston, MA, USA

Applied AI Engineer

Sep 2024 – June 2025

- Created **10+ AI applications** from scratch, building custom **RAG pipelines** and **multi-agent systems** without relying on orchestration frameworks.
- Constructed end-to-end RAG pipelines for patient history retrieval, clinical triage, and patient routing by implementing document ingestion, embedding generation, semantic search, retrieval, reranking, prompt engineering, and prompt orchestration across knowledge bases containing **10K+ documents**.
- Orchestrated agentic AI systems incorporating **planning, memory, tool orchestration, multi-hop reasoning, debate, routing, evaluation, and autonomous decision-making** workflows for complex clinical decision-support tasks.
- Benchmarked transformer-based language models, embedding models, and vector database implementations (**FAISS** and **ChromaDB**) to optimize semantic retrieval quality, inference cost, and deployment performance for low-resource healthcare environments.
- Enabled local (**Ollama**) and cloud-based (**Groq**) inference services, allowing production AI applications to operate across environments with varying connectivity and compute constraints.

AFTERQUERY EXPERTS

San Francisco, CA, USA

Software Engineer - Platform Reliability & Tooling

Mar 2025 – May 2025

- Strengthened the reliability of Python-based ML and data processing pipelines by debugging production software issues across **NumPy, Pandas, and PyTorch**, reducing runtime failures by **40%**.
- Expanded automated test coverage from **68% to 92%** by developing end-to-end testing workflows, improving deployment reliability and reducing onboarding effort.

IG GROUP

Data Science Intern

Bangalore, India

Aug 2021 – Nov 2021

- Engineered **XGBoost-based fraud detection models** for high-frequency financial transactions, supporting fraud prevention initiatives that reduced estimated losses by **₹12L per month**.
- Optimized transformer-based semantic search pipelines that improved retrieval of historical fraud cases, increasing analyst productivity by **45%**.
- Automated ETL workflows using Apache Airflow, eliminating **10+ hours/week** of manual reporting while improving downstream data availability.

INDIAN SPACE RESEARCH ORGANIZATION(ISRO)

Project Trainee

Bangalore, India

Mar 2021 – Jun 2021

- Developed predictive monitoring models using **10+ years** of telemetry data to detect server overheating risks, enabling proactive anomaly detection through monitoring dashboards and alerting workflows that reduced response time by **30%**.

EDUCATION

NORTHEASTERN UNIVERSITY

Master of Science in Artificial Intelligence - 3.42/4

Boston, MA

Jun 2024

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Bachelor of Engineering in Computer Science - 8.77/10

Bangalore, India

Jul 2022

PROJECTS

MULTI-AGENT AI REASONING PLATFORM

- Developed an extensible **multi-agent reasoning framework** integrating **SQL databases, REST APIs, and document retrieval pipelines** using Groq and Ollama.
- Reduced complex business query response time to **under 10 seconds** through staged reasoning and parallel tool execution.

EXPLAINABLE NBA GAME PREDICTIONS

- Developed an NBA game outcome prediction system achieving **85% accuracy** using **XGBoost** with feature engineering and hyperparameter tuning, while leveraging **BERT, Llama, and Mistral** for natural-language explanation and analysis of model predictions.
- Applied **SHAP** and **LIME** to explain model predictions, visualizing feature importance and decision behavior through Matplotlib and Seaborn to improve model interpretability.